

electron/prefetch-summaries.cjs

Kind: added · Baseline: a9c34bfb03fe · Target: NovaDiff · Generated: 2026-05-20T04:30:24.831Z

Overview

A new module `electron/prefetch-summaries.cjs` (lines 1-241) adds background pre-fetching of LLM-generated file summaries. It orchestrates a Rust CLI queue via `runCompareEngine(..., cmd:"prefetch-summary-queue")` and streams progress through IPC events `summary-prefetch-progress`.

Key changes

- **Imports** (lines 9-12): `runCompareEngine`, `summarizeChange`, `buildDiffExcerpt*`, `exportFileSummaryArtifacts`.
- **Cache & abort**: `summaryByPath` (Map, line 14) and `prefetchAbort` (AbortController, line 17). Functions `clearSummaryPrefetchCache()` (lines 20-22) and `stopSummaryPrefetchWorker()` (lines 24-28) manage state.
- **Prefetch loop** (`runSummaryPrefetchLoop`, lines 64-216): builds a job queue, iterates over jobs, obtains a diff payload, builds chunks, calls `summarizeChange`, stores the result, optionally exports artifacts, and emits IPC states (`started`, `file-done`, `file-skipped`, `file-error`, `finished`).
- **Worker control** (`startSummaryPrefetchWorker`, lines 222-232): starts the loop with a new `AbortController`, returns a promise that resolves on completion.
- **Default limit** (`defaultPrefetchLimit`, lines 39-47): reads `NOVADIFF_PREFETCH_MAX`, clamps 1-5000, defaults to 200.
- **Exports** (lines 235-241): all public helpers.

Impact

- **Performance**: Background worker may increase CPU/memory during large diff sets; aborting mitigates runaway jobs.
- **Observability**: `summary-prefetch-progress` IPC channel provides fine-grained progress for UI/debugging.
- **Configuration**: `NOVADIFF_PREFETCH_MAX` controls queue size.
- **Packaging**: Additional JS dependencies and exported artifacts increase bundle size; consider lazy loading.
- **Error handling**: Errors are caught and reported via IPC, preventing crashes but requiring UI handling of `file-error`.

Risks & follow-ups

- **Race conditions**: Verify `prefetchAbort` cancels in-flight jobs and `stopSummaryPrefetchWorker` cleans the controller.

- **Cache leakage:** Ensure `clearSummaryPrefetchCache()` runs on worker restart and app shutdown.
- **IPC reliability:** Test that all `summary-prefetch-progress` messages reach the renderer, especially when `webContents` is null.
- **Environment variable parsing:** Confirm non-numeric or negative values fall back to 200 and clamping to 5000 works as intended.